RMIT Classification: Trusted

MANU2476 Automated System Design

SCADA System Design

(Max 25 pages)

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What's next...

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1. Introduction (15 pts)

The project involves developing control software for an automated material handling system intended to sort and store workpieces. The system must operate under different conditions, allowing operators to initiate single or continuous processing modes. The goal is to ensure the system processes items swiftly, completing each cycle in less than 8 seconds while providing real-time status updates and control functions via the HMI.

1st System Configuration

At the beginning the user will turn on the PLC, the HMI, and the air compressor. After that, wait few seconds for the hardware and electricity to synchronize. Then the user needs to download the control program from the programming device to the PLC. The work continues with transferring the HMI configuration to the HMI device to allow monitoring and control.

2nd System Initialization

The operation is initiated when the operator presses the Start command, either via the HMI interface or the physical Start button on Module C. This action does not trigger the entire process immediately; instead, the system prompts the user to select the desired operating mode. The operator must choose between Mode 1 (single cycle), where the system executes only one handling sequence before stopping, or Mode 2 (continuous cycle), where the process runs continuously until a Stop or Fault button is issued.

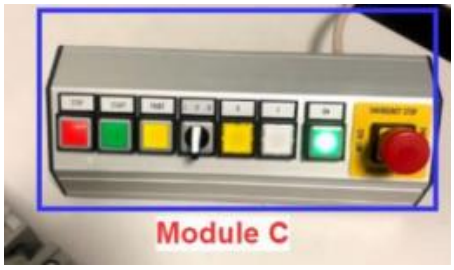


Figure : Module C

3rd Workpiece Loading – Station 1 (Module A)

Upon receiving the execution command from the controller, the sensor located beneath the magazine checks the presence of a workpiece. If a workpiece is available, the pneumatic cylinder in this module advances one workpiece at a time from the magazine into the inspection area. Within this area, dedicated sensors identify the material type and color — distinguishing between

metal and plastic, as well as differentiating black plastic from white plastic. If no workpiece is detected in the magazine, the system generates a fault signal and halts operation.

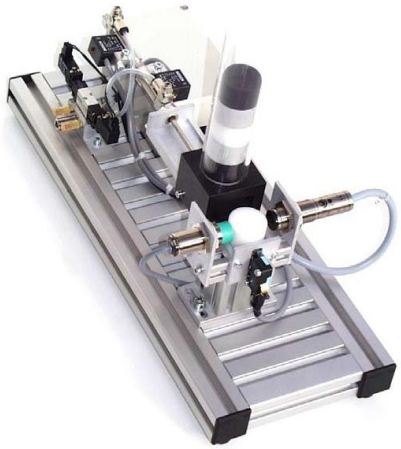


Figure : Module A

4th Processing Transfer Unit – Station 2 (Module B)

The program in Module B primarily laid on the Robot Arm and the Vacuum suction cup attracts workpiece. So at the beginning of the program, robot arm will be at the Middle. When the workpiece at module A is pushed, the Workpiece will touch the Mechanical limit switch, this is the key to activate the robot arm to move immediately to Module A to attract the workpiece to itself, after 1-2 seconds, it will start going to Station 4 – Module D.

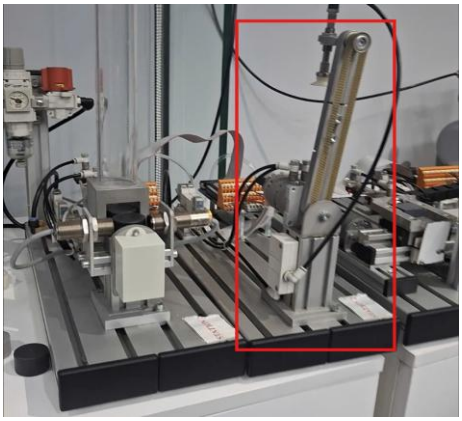


Figure : Robot arm of Module B is at middle at the initialization step

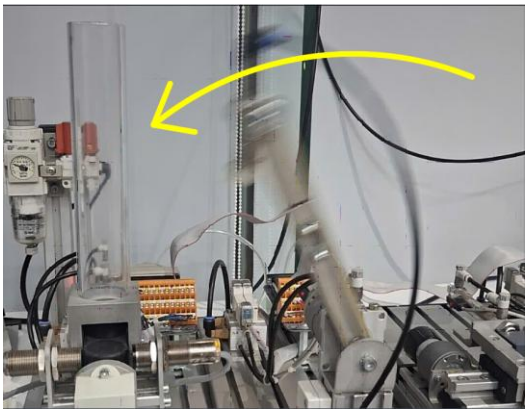


Figure : Robot arm moves to Module A

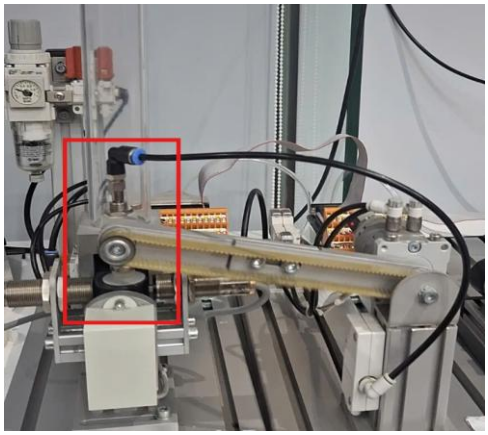


Figure : Robot arm attracts workpiece

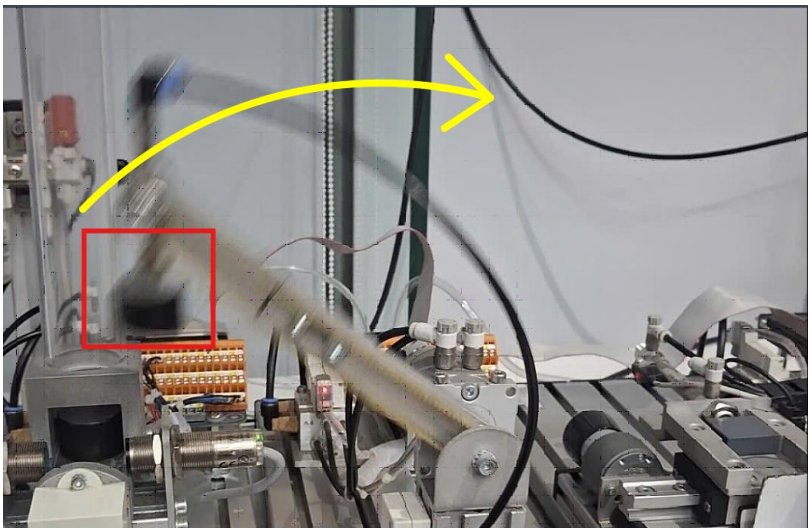


Figure : Robot arm moves to Module D with workpiece attracting

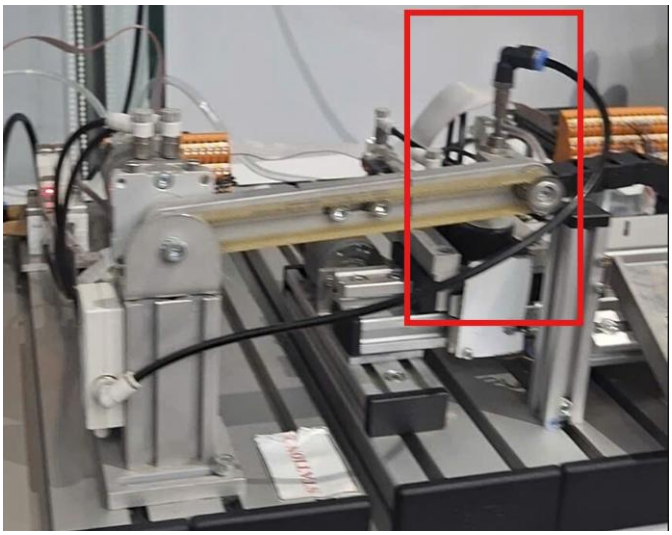


Figure : Robot arm reaches Module D with workpiece

5th Tray Slider Storage Unit – Station 4 (Module D)

When a workpiece arrives at Station 4, the conveyor synchronizes with the tray slider, moving it to the right until it aligns with the designated slot for that specific type of workpiece. For instance, Slot 1 is assigned to black plastic workpieces, Slot 2 to white plastic workpieces, and Slot 3 to metal workpieces. Once the slider reaches the correct position, the pneumatic cylinder mounted on the slider pushes the workpiece into the slot. After placement, the cylinder retracts to its initial position, and the conveyor returns to the left, ready to receive the next workpiece.

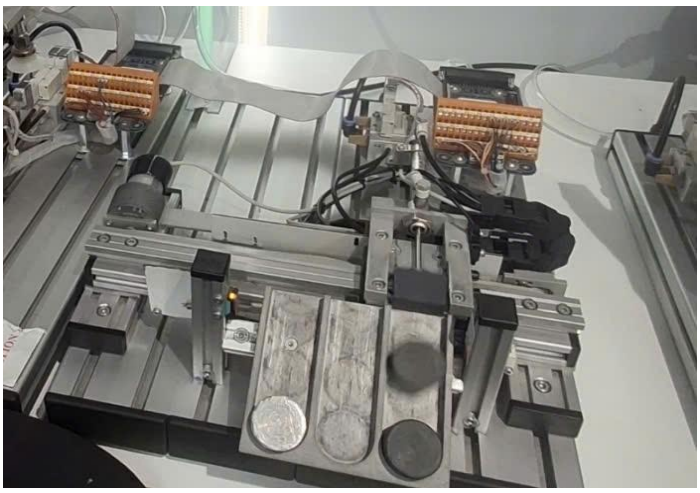


Figure : The cylinder on the Tray Slider of Module D is pushing a black workpiece in the third slot

The project involves designing a software system to control an automated material handling process, capable of operating in both single and continuous modes, with real-time monitoring and control provided via an HMI. To achieve these requirements, development of a PLC-based control program

that will manage the sequence of operations which is based on the work flow the whole system, including sensors and actuators. The software will implement dual operating modes with conditional logic to switch between single and continuous cycles based on user input from the HMI. The HMI will be designed to display information such as the system status, process count, while also providing intuitive control buttons for start, stop, and modes.

2. Module Explanation

2.1. Module C

- What are the main objectives of the Module C?

The main objectives of the Module C are to provide a manual control interface for starting, stopping, and resetting the automated system. The module allows the selection of different operation modes. In addition, it also includes the emergency stop function to ensure safe operation.

- What is the main operational principle of the Module C?

The main operational principle of the Module C is to send logical input signals to the PLC and the other modules operate based on the program. The program starts to run when the start button is pressed, and the operation mode is selected by the switch that the switch I selects mode 1 (single cycle) and the switch II selects mode 2 (continuous cycle). The fault mode is triggered for some specific conditions such as an empty magazine or full sorting tray. It allows the other modules to go back to the beginning setting.

- What are buttons and their function of the Module C?

The module C contains seven buttons and switch, but not all buttons are used for this system. The start button is used to initialize the process and activates the system. The stop button is used to stop the process immediately. The two mode selection switch allows the user to choose between single cycle mode and continuous cycle mode. The fault button is used to trigger the fault mode. For pressing fault button one time, the module B gets reset and the cylinder rotates back to the original position (middle of left and right sensor). Pressing the fault button twice allows the conveyor on the module D to go to the left limit.

2.2. Module A

- What are the main objectives of the Module A?

The main objectives of the Module A are storing workpieces in the magazine and detects the color/type of workpieces.

- What is the main operational principle of the Module A?

If there is workpiece in Module A, the piston pushes the workpieces forward for detecting the type of workpiece and let Module B grabs it, as the workpiece is ready, piston retracts, and the detected information is stored to sort them out in Module D later.

- What are the I/Os and their function of the Module C?

The inputs of the module A are the sensor for the cylinder to check if it has retracted or extended, sensor for magazine to check if there is workpiece or the magazine is empty, sensor for workpiece to check if the workpiece is ready for pick up by the module B, and two more sensors to detect the workpiece is whether white or black and metal or plastic. The outputs of the module A are cylinder to be extended and retracted for moving the workpiece from magazine.

2.3. Module B

- What are the main objectives of the Module B?

The main objective of the Module B is delivering workpieces from module A to module D.

- What is the main operational principle of the Module B?

As the workpiece ready on the module A, the robot arm rotates left and stays for a while, making sure that the vacuum is on the workpiece, then the vacuum sucks the workpiece up and delivers the workpiece to the conveyer on the module D. When it successfully delivers the workpiece to the conveyor, the vacuum releases workpiece, the robot arm rotates left a bit and stops, returning to its original position, at the middle where sensor left and sensor right of the arm are off, waiting for the module D completes sorting workpiece and another workpiece ready on the module A, ready for another workpiece.

- What are the I/Os and their function of the Module C?

Module B has two input sensors detecting whether the robot arm is at right or left, the outputs of Module B make robot arm rotate left or right and control the vacuum's suction system.

2.4. Module D

- What are the main objectives of the Module D?

The main objective of Module D is sorting out workpieces.

- What is the main operational principle of the Module D?

After receiving workpiece delivered by the module B, the conveyor sorts the workpiece base on its color and type detected in Module A, the sorting principle works as follow: As the tray moves right to sort workpiece, the U-Shape detector counts how many U-Shape it has encountered, if it detects 1,

2 or 3 then the conveyor stops and push workpiece in socket then retracts piston (according to color of workpiece, white is 2, black is 1 and metal is 3). After that, the conveyor moves left to its home position (left limit position), ready for another workpiece.

- What are the I/Os and their function of the Module C?

For the inputs of the module D, it has two sensors on each side of the conveyor track. They are used to detect the position of the conveyor. The u-shape sensor on the workpiece holder will detect the location of the workpiece sorting slot. The optic sensor on the side of workpiece sorting slot will detect whether the slot is full or not. The outputs of the module D are the conveyor motions that either the conveyor turns left or right, and the cylinder extension to deliver the workpiece to the sorting slot.

2.5. Module E: HMI

- What are the main objectives of the Module E?

The main objectives of Module E is to provide a GUI (Graphical User Interface) to control the whole system instead of manually pressing physical button like Module C, acts as an alternative of Module C.

- What is the main operational principle of the Module E?

Module E (HMI) opens to Screen 1 with ‘Start’ and ‘Stop’ mirrors of Module C plus a ‘Mode’ key and a yellow box shows working status (counts how many workpieces have been sorted so far in the process). Tap ‘Mode’ to reach Screen 2, offering four choices: Main (return to Screen 1), Mode 1 (single cycle), Mode 2 (continuous run), and Fault (emergency stop).

- What type of communication to connect the PLC and HMI?

The program is compiled and downloaded to the S7-1500 and runs the modules while the HMI program is uploaded to S7-1200 and connects it to the S7-1500 PLC since S7-1500 does not have display to show the HMI. The connection is made by a PROFINET cable.

3. Software Algorithm and Design

3.1. Structured Programs (10 pts)

In this section, you need to provide a flow chart to represent the overall operation of the system and then provide the main structure of your control program/software.

3.2. Network Analysis and Explanation (30 pts)

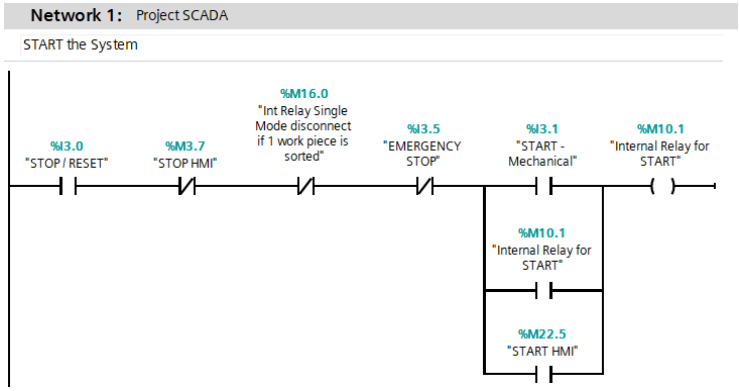


Figure : Network 1

In network 1 of the Main OB1, input I3.1 is the START physical button and M22.5 is the START from the HMI, pressing one of them will activate the M10.1 (the internal relay for the START). If the operator wants to stop the program, press either I3.0 – STOP physical button or M3.7 STOP from HMI. The normal close M16.0 was designed to act as a STOP for the single mode, if the workpiece in the single mode completely delivers into the slot, it will activate M16.0 to stop the program.

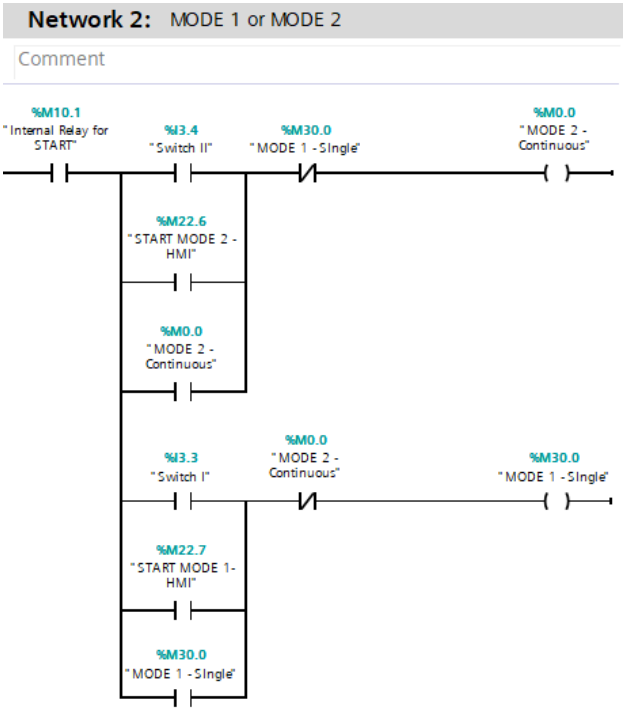


Figure : Network 2

In network 2, after M10.1 is activated from network 1; the user needs to choose single mode or continuous mode to perform. If the user wants to perform continuous mode, switching the I3.4 (Switch II) or press M22.6 (Start Mode 2 from HMI). If the user wants to perform single mode,

switching the I3.3 (Switch I) or press M22.7 (Start Mode 1 from HMI). The outputs are internal relay for continuous mode M0.0 and single mode M30.0 respectively. The 2 normal closes are designed have the same address as the output for the safety interlock purpose.

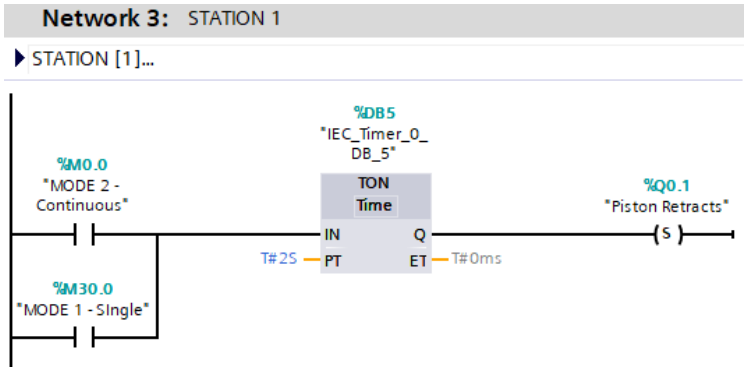


Figure : Network 3

In network 3, both internal relay of single mode and continuous mode will enter the Timer which wait 2 seconds initially then set retracts the cylinder Q0.1 of the Module A.

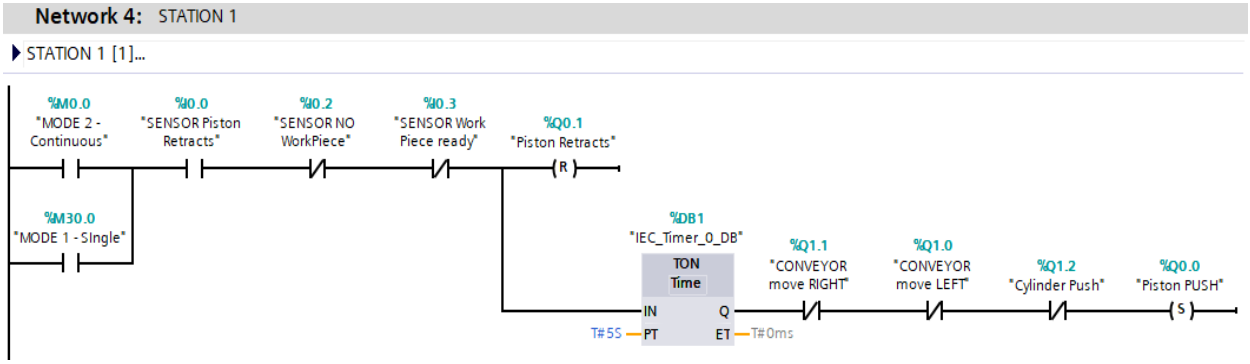


Figure : Network 4

In network 4, both modes will also activate this network. The cylinder of station 1 retracts until it triggers the sensor retracts I0.0. It allows the program continuously run into the reset output Q0.1. After 5 seconds, reset the retracts, the Timer will allow the set output pushing the workpiece Q0.0. The normal closes in this network also act as the safety interlock. For example, if the conveyor is running no matter left or right, the cylinder on the module A is not allowed to push. This helps the program to wait for 1 workpiece to finish in order to start the next cycle.

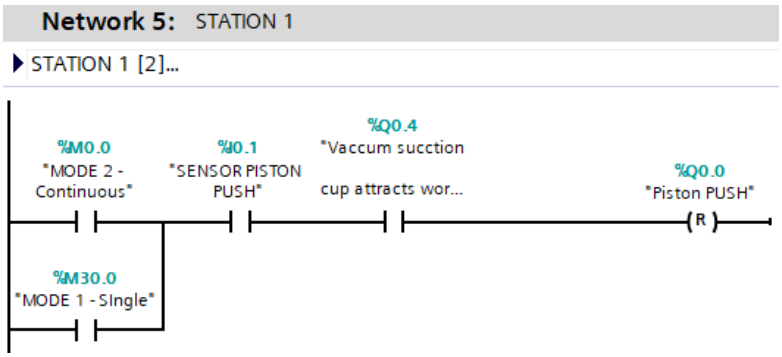


Figure : Network 5

In network 5, the cylinder push into it meets the sensor push I0.1. It waits until the vaccum of module B to attract, then it goes to output reset the push Q0.0.

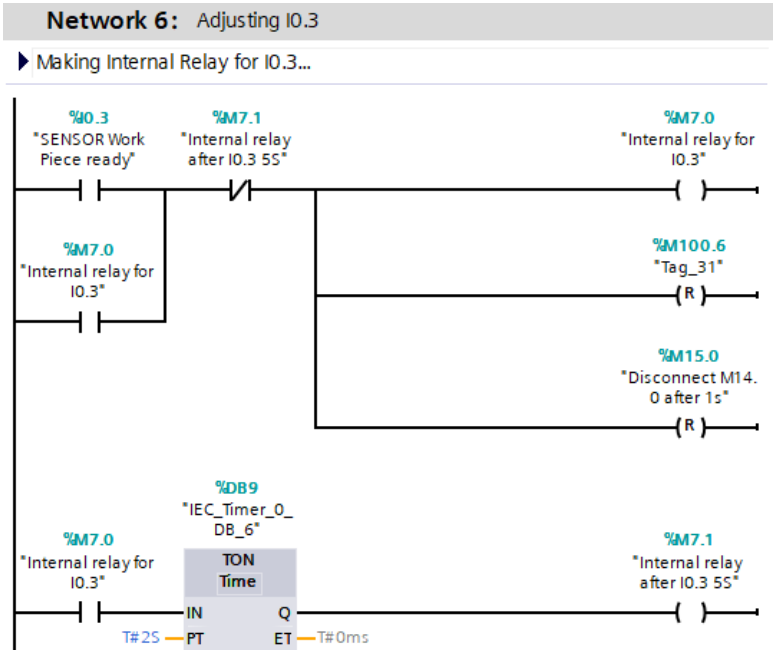


Figure : Network 6

In network 6, since the mechanical limit switch I0.3 is used to detect if the workpiece has been pushed or not only be activated for a short period. This network is a solution for it. The I0.3 when activate will also activate the M7.0, this internal relay will be used to replace I0.3 for 2 seconds. It means it will lengthen the existing time of I0.3 for 2 seconds. After 2 seconds M7.1 will disconnect M7.0.

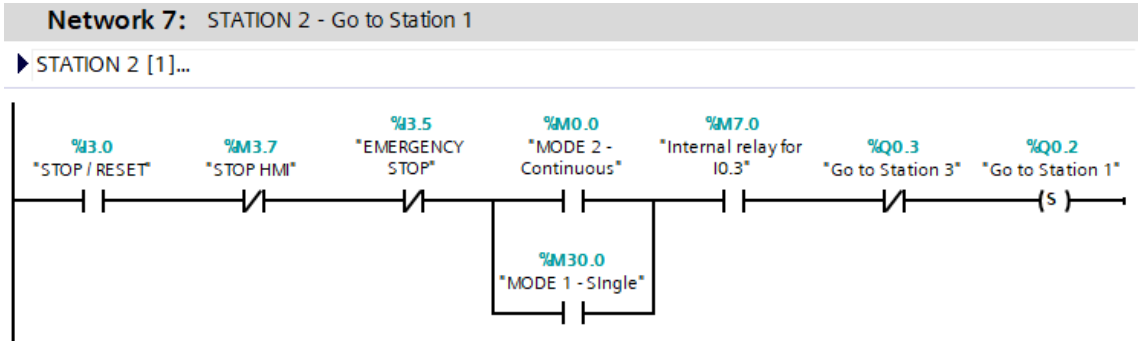


Figure: Network 7

In network 7, both modes will activate the output go to the left of robot arm of station 2 – Module B. Thanks for the help of M7.0, this allow enough time for the electricity go through without disrupted.

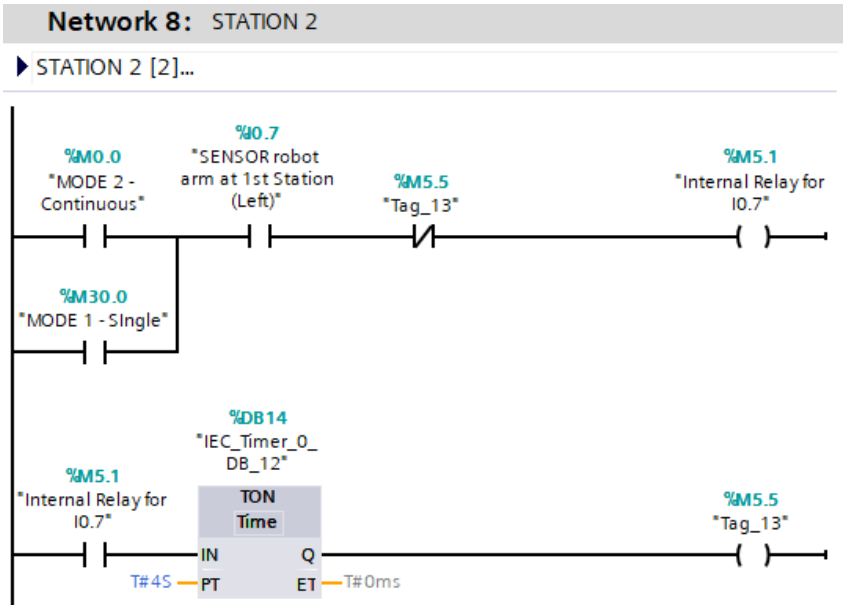


Figure 8: Network 8

In network 8, both modes will also trigger if the robot arm move to station 1. It means the sensor on the left I0.7 is activated. Making internal relay for I0.7 is also important, because of error in the mechanical, I0.7 is sometimes appears just for short time. M5.1 will replace I0.7 for a short period 4 seconds before interrupted by M5.5.

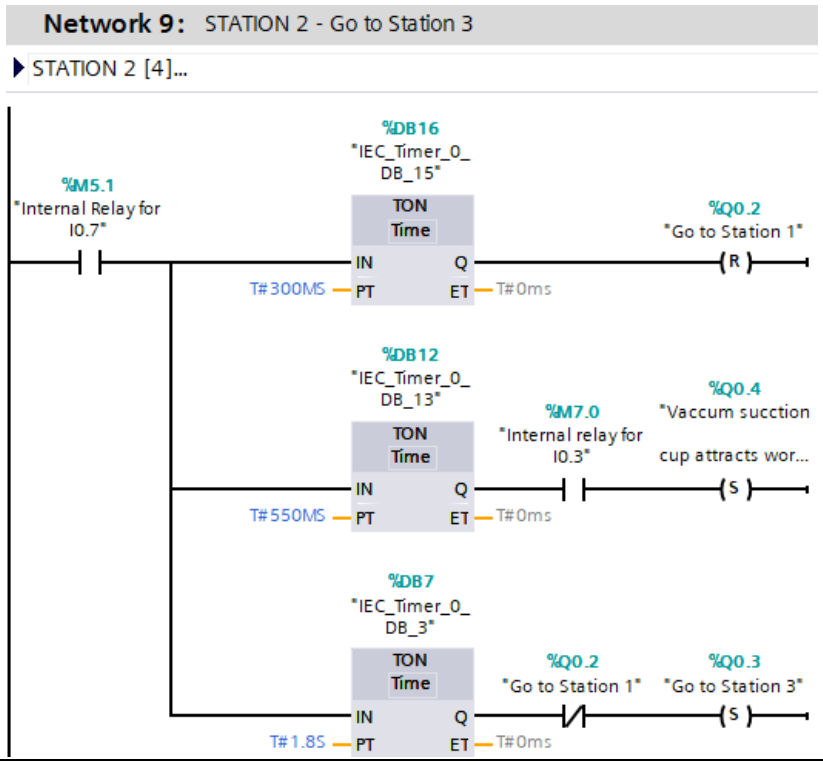


Figure 9: Network 9

In network 9 after internal relay M5.1 is closed, it will start to run each timer DB16, DB12, DB7. For 550ms the suction of the arm is on and keep the piece to travel to station 3. For 300ms, the Q0.2 will be reset for another piece to trigger it to take again. For 1,8s the robot arm will travel from station 1to 3 and put the piece on the tray.

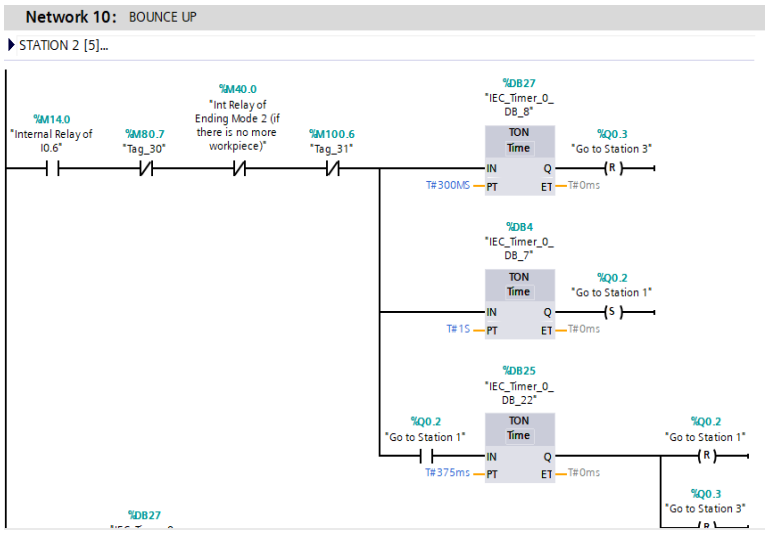


Figure : Network 10 [1]

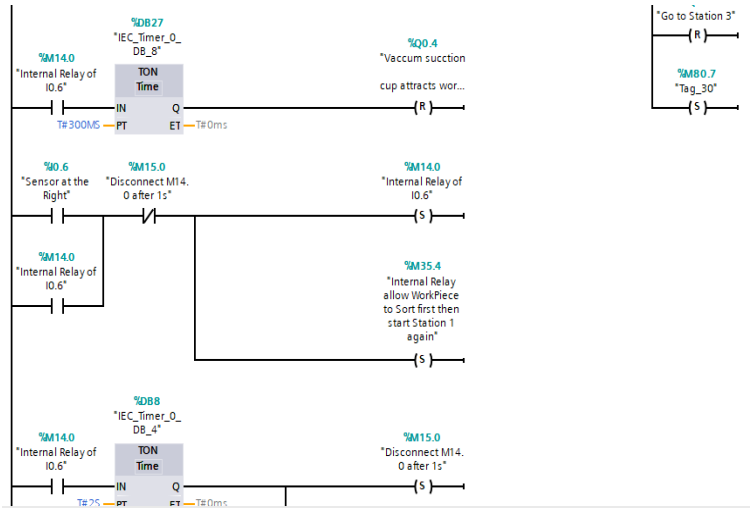


Figure : Network 10 [2]

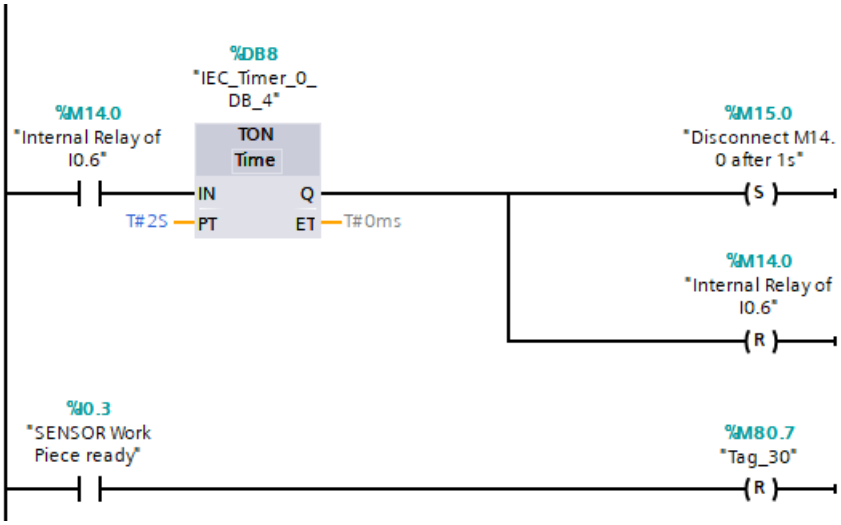


Figure : Network 10 [3]

This network 10[1] is responsible for the bounce of the robotic arm. The first timer, DB27, is enabled by a combination of internal relays and a normally closed contact, which acts as a check to ensure there are still workpieces to process. After a 3-second delay, this timer's output activates the Q0.3 to reset it. The network also contains DB4, which controls the command Q0.2 for return to station1 for 1s or “bounce” and DB25 immediately reset both of Q0.2 and Q0.3, stopping the arm to travel all the way to station 1 and set M80.7

This 10[2] manages the vacuum suction reset. When this timer is complete, it also resets an internal flag M80.7. The middle uses another TON timer DB6, triggered by a sensor, to activate a relay M15.0 after a 1-second delay, and another relay M35.4 that allows the sorting and restarting of the process at Station 1.

This 10[3] focuses on resetting the main control relay. It features a TON timer DB8 with a 2-second preset time. This timer begins counting when the internal relay M14.0 is true. After the 2-second delay, the timer's output performs two functions: it sets the M15.0 relay and resets the %M14.0 relay. This action of resetting M14.0 turns off the timer's input, causing the entire sequence to reset and prepare for the next cycle. For I0.3 ("SENSOR Work Piece ready"), provides a path to reset the M80.7 relay.

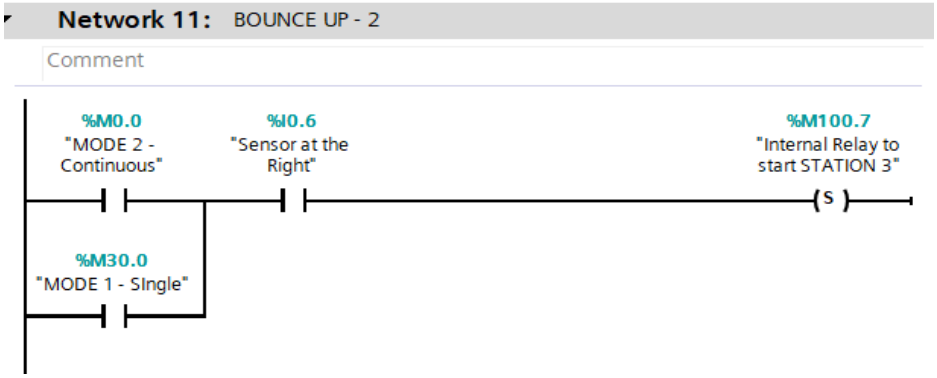


Figure : Network 11

This network when any mode turns on with the sensor at right, it will trigger M100.7 relay. This is used to start the Stray Slider module.

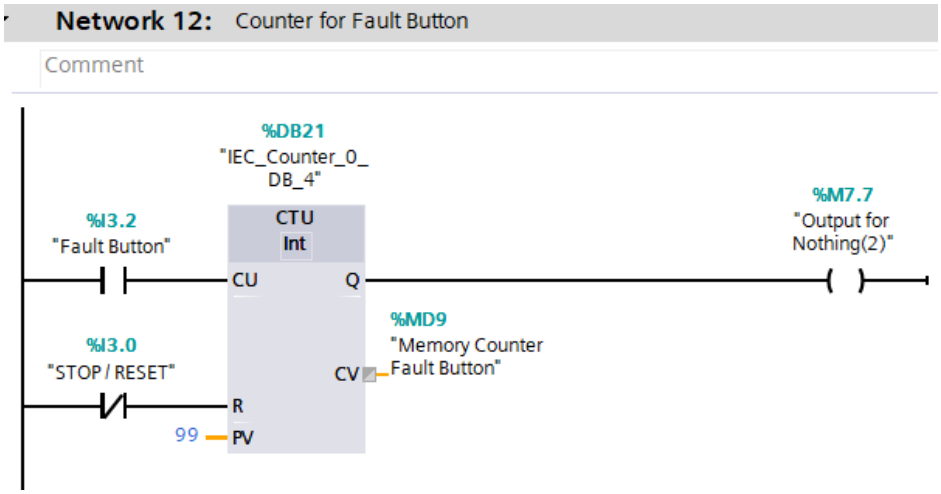


Figure: Network 12

This network is for fault button. After the fault button is pressed, it will count to 1 and sent to M7.7. The I3.0 is the reset button for the counter.

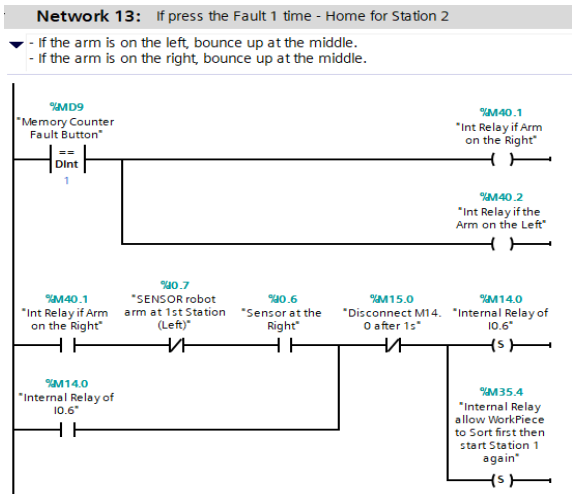


Figure : Network 13 – Press Fault Button 1 time [1]

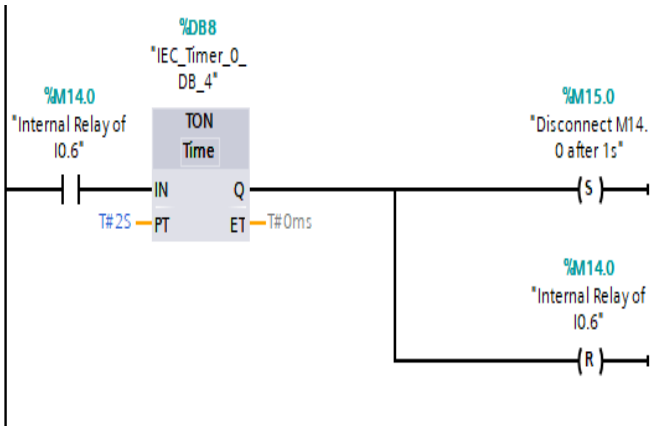


Figure : Network 13 – Press Fault Button 1 time [2]

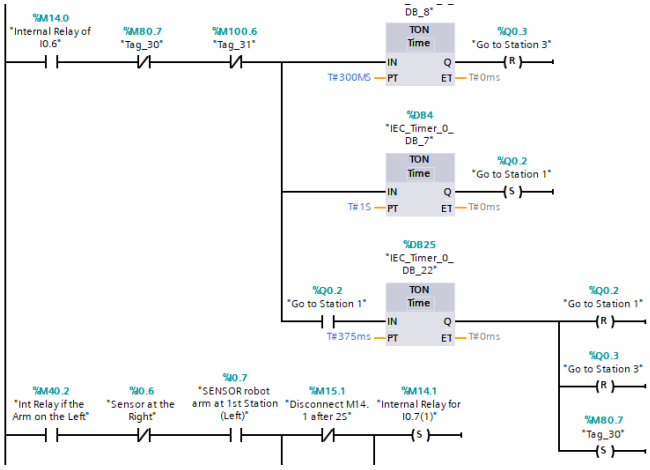


Figure : Network 13 – Press fault button 1 time [3]

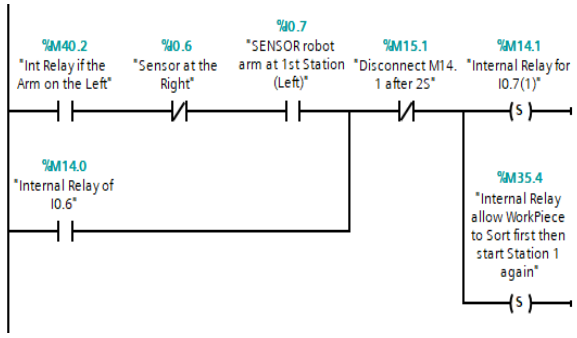


Figure : Network 13 - Press fault button 1 time [4]

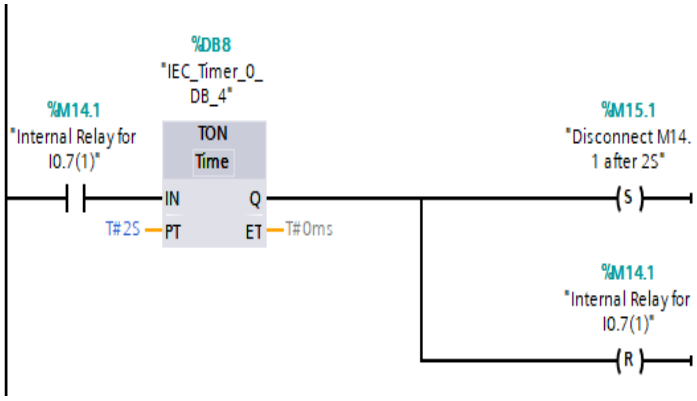


Figure : Network 13 - Press fault button 1 time [5]

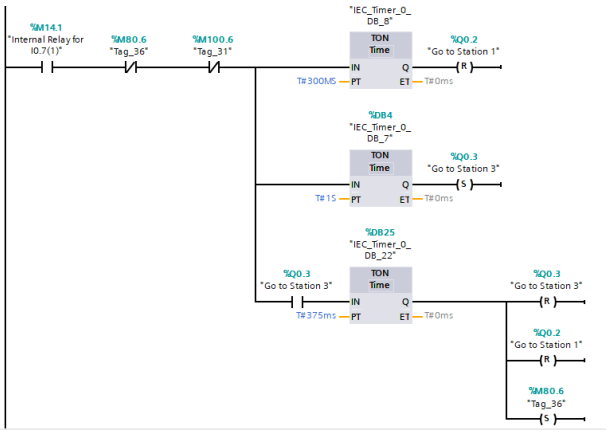


Figure : Network 13 - Press fault button 1 time [6]

In network 13, if the the Fault button pressed 1 time, it will trigger the output M40.1 (Internal relay if robot arm on the right) and M40.2 (Internal relay if robot arm on the left). If the robot arm on the right, it triggers sensor I0.6, setting M14.0 to exist for 2 seconds. Then the rest work the same as the

bounce up section. If the robot arm on the left, it would be similar, but instead of triggering M14.0, it triggers M14.1 to bounce up to the right and reset while in process to stop at the middle. This would solve the scenarios of the robot arm at the beginning if it is not at the middle but somewhere on the left or right.

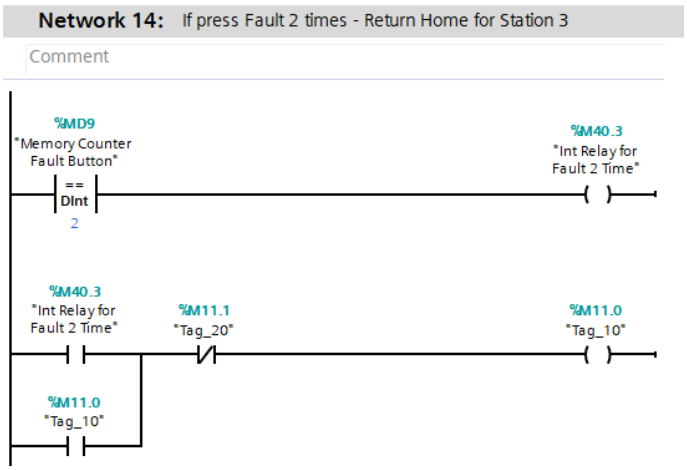


Figure : Network 14 – Press fault button 2 times [1]

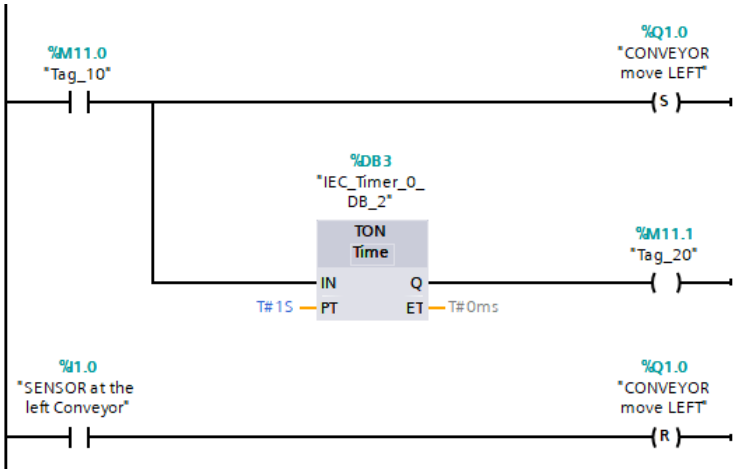


Figure : Network 14 – Press fault button 2 times [2]

This network is is when the button is pressed 2 times it will trigger the M40.3 which keeps the M11.0 and set the convey to move left Q1.0 and start the timer to trigger M11.1 for 1s to turn the M11.0 for 1s. When I1.0 is on, Q1.0 is reset. This will make conveyor whenever the position it stays at the beginning always go back to the left (home position).

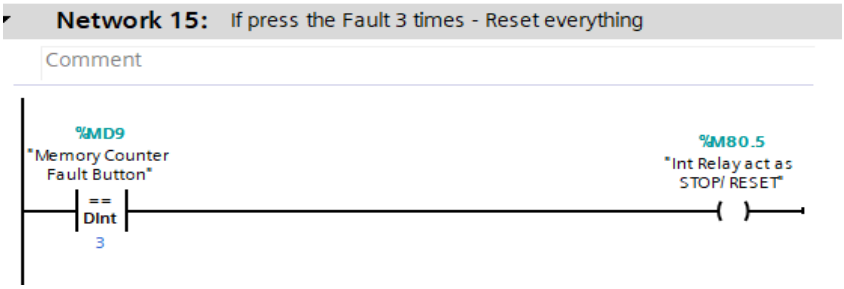


Figure : Network 15 – Press Fault button 3 time

This network is when the fault button has been pressed 3 times it will trigger M80.5. This M80.5 is linked to network 16 and network 17. It will reset everything in the program, act as a stop button.

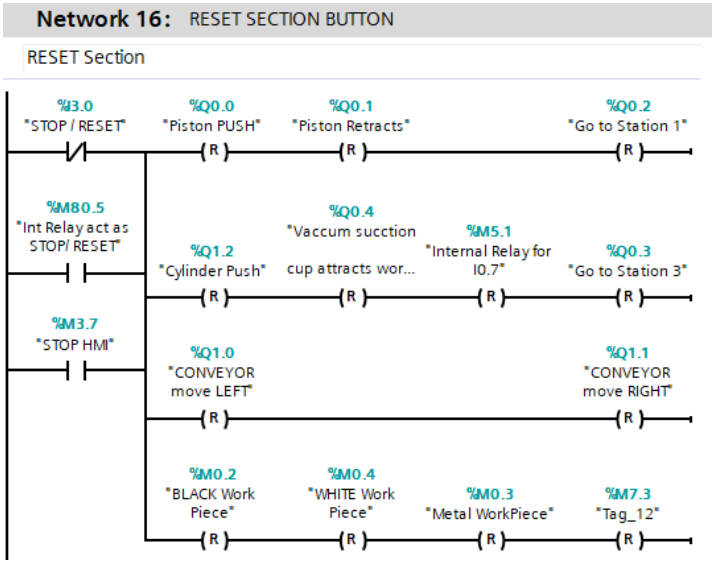


Figure : Network 16 – Reset section button [1]

This network is for reset all the output (Q0.0, Q0.1, Q0.2, Q1.2, Q0.4, Q0.3, Q1.0, Q1.1) and all the relays. The M80.5 act as a signal to reset all the outputs, the M3.7 act as a tag for HMI reset/stop.

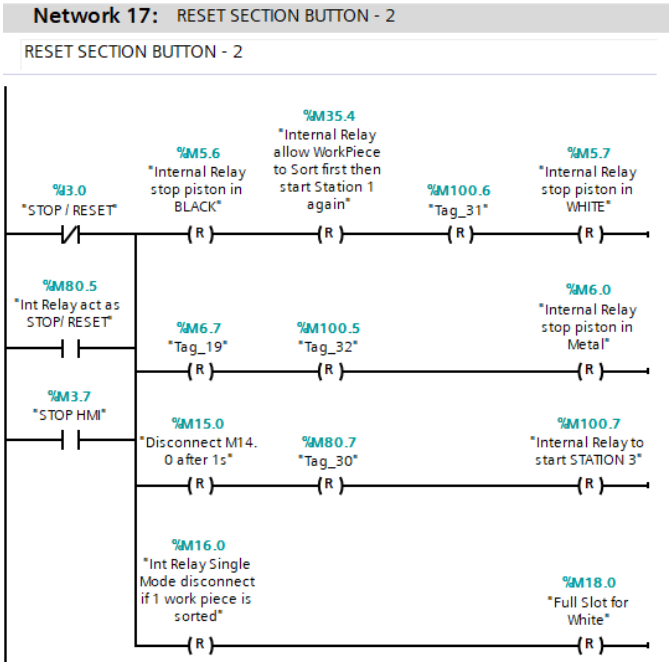


Figure : Network 17 – Reset section button [2]

Reset everything as network 16.

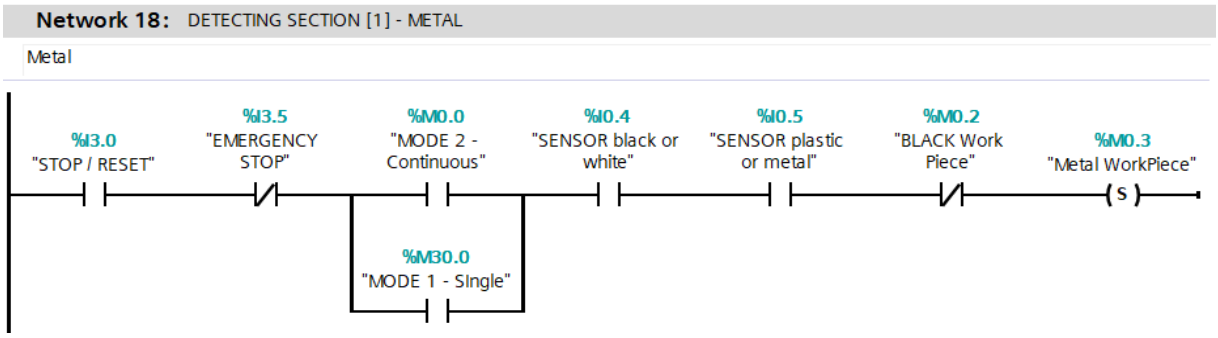


Figure : Network 18 – Detecting for Metal

In network 18, this is where the 2 sensors in module A detect metal workpiece. According to the logic on the table below, the 2 sensors determine the type and color of workpiece. (network 19, 20 are using same methods for detecting white and black workpiece - appendix).

	I0.4	I0.5
Black	1	0
White	0	0
Metal	1	1

Table : Logic to detect Black, White and Metal of I0.4 and I0.5

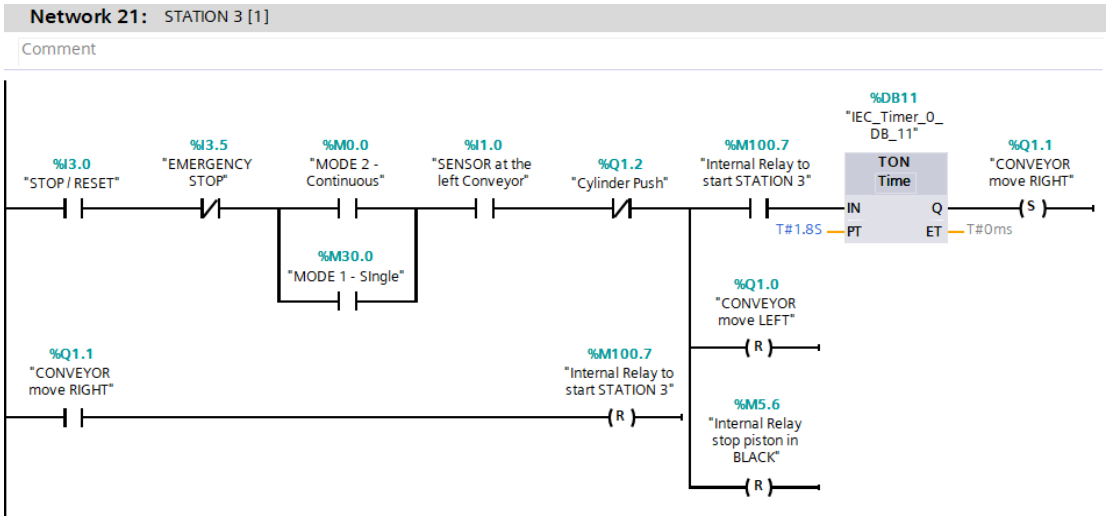


Figure : Network 21 – Station 4 [1]

When the station 2 successfully delivers the workpiece to the station 4, the right sensor for the robotic arm turns on and the conveyor is at the left limit to receive the workpiece, then the convey moves to right after 1.8 seconds by the timer.

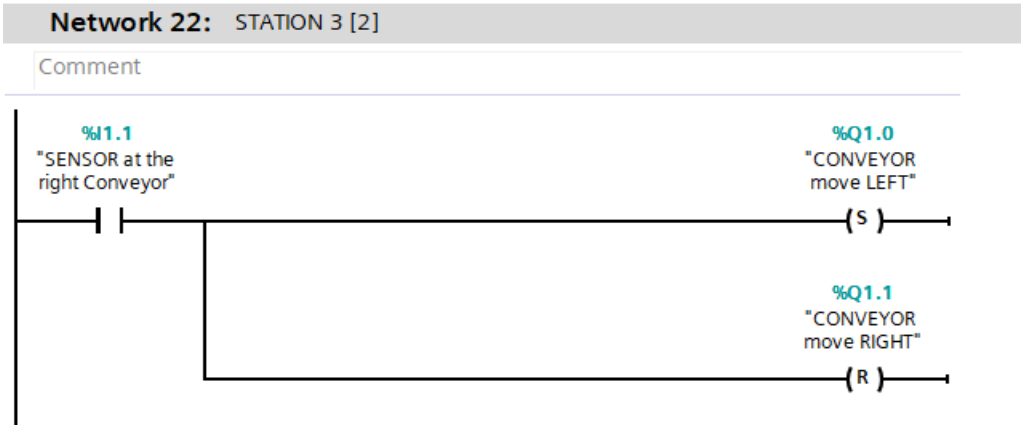


Figure : Network 22 – Station 4 [2]

When the conveyer at the right limit, it starts to return back to home position.

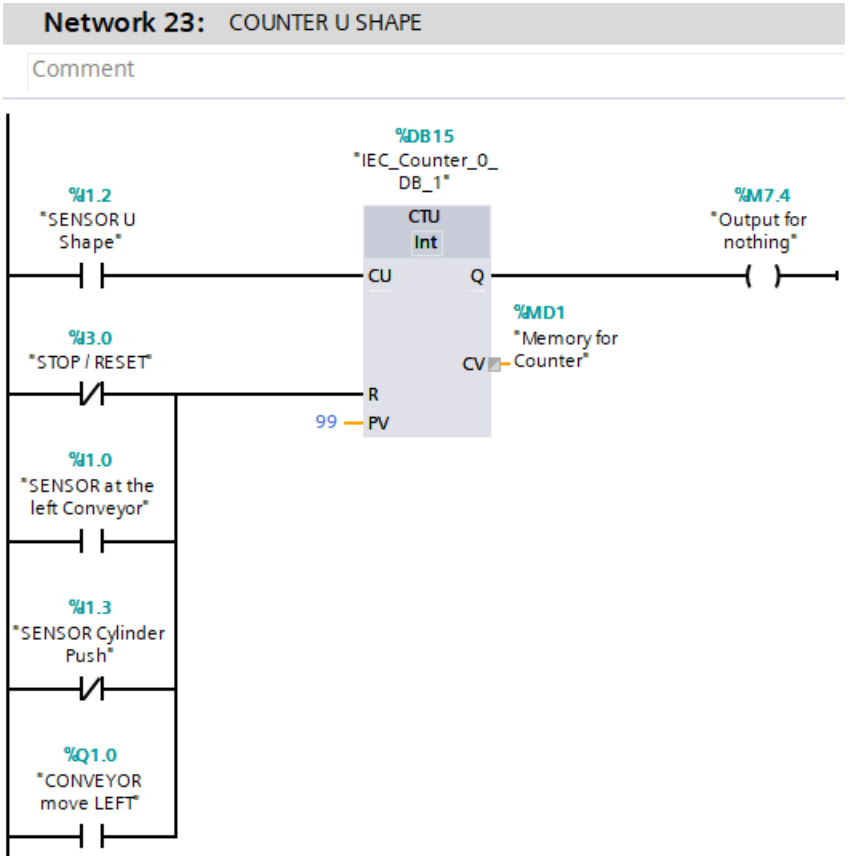


Figure : Network 23 – Counter U Shape for Station 4

Count how many U-Shape the conveyer has passed to determine which socket the workpiece belongs to, As the conveyer at the leftmost / workpiece is pushed to socket / conveyer moving left to home position, the counter reset / does not count.

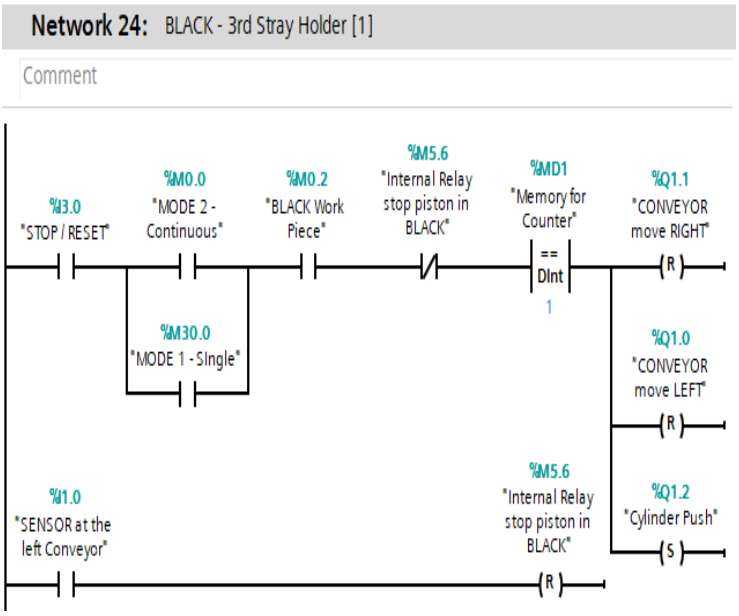


Figure : Network 24 – Sorting Black [1]

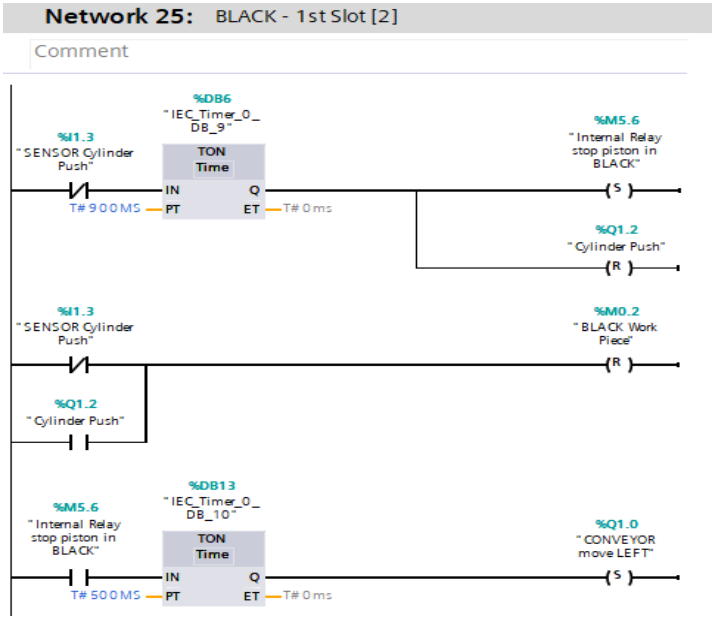


Figure : Network 25 – Sorting Black [3]

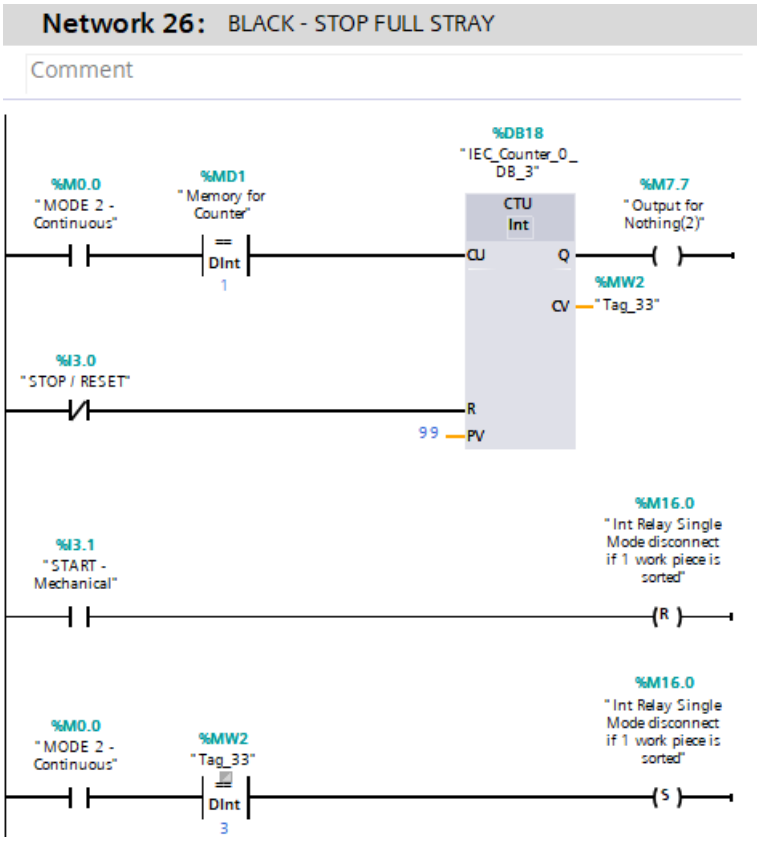


Figure : Network 26 - Full slot scenario for Black Workpiece

In Networks 24, 25, and 26, memory bit MD1, linked to the counter in Network 23, is activated when the detected workpiece is black and it is the first item in the sequence. At this point, the conveyor moves the tray slider to the first slot position, and the pneumatic cylinder (Q1.2) pushes the black workpiece into the slot. After the push action, the cylinder retracts, and the sensor confirms its return. This feedback resets the push sequence and triggers internal relay M5.6, which then commands the conveyor to return to the left after a 500ms delay.

In Network 26, if the incoming workpiece is black, the counter increments in MW2. When the counter reaches a value of 3, indicating that the black workpiece slot is full, the comparator output sets M16.0. This bit functions as a stop condition, terminating the process for single mode operation and preventing further black workpieces from being inserted into an already full slot.

White and metal have the same method of sorting. It can be seen in the appendix from network 27 to network 29 for Metal and network 30 to network 32 for White.

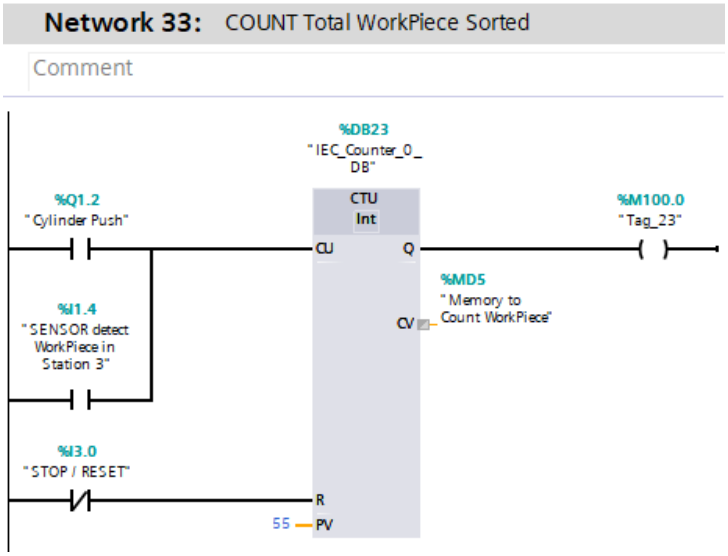


Figure: Network 33 – Counter counts total workpiece sorted

In Network 33, each time the cylinder (Q1.2) performs a push action or when the sensor (I1.4) detects the presence of a workpiece at Station 3, the counter (DB23) increments by one. The accumulated count is stored in memory location MD5. This value is then linked to the HMI, enabling the operator to continuously monitor the total number of workpieces that have been successfully sorted during the process. Additionally, the counter can be reset to zero by pressing the Stop/Reset input (I3.0), allowing a new counting cycle to begin.

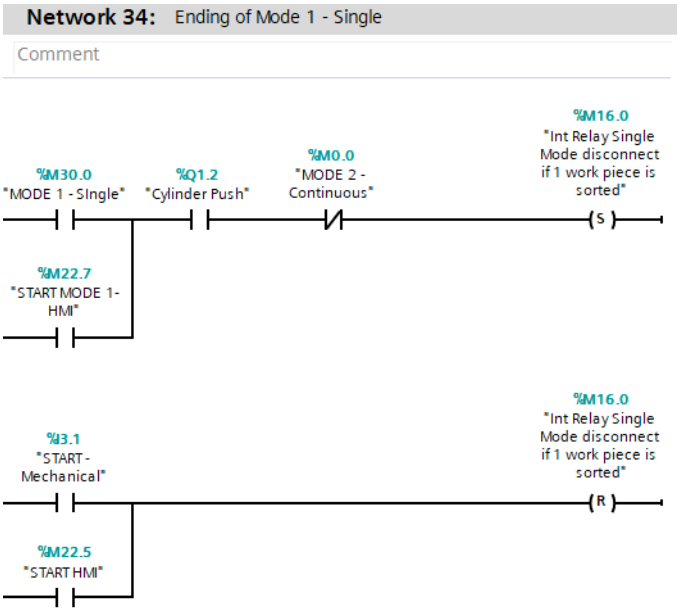


Figure: Network 34 – Ending of Mode 1

This network is the ending mode of single mode (Mode 1), it can be activated either from the HMI or physical button, then wait until a cylinder output is pushed, go to output M16.0 to set to disconnect the program. This M16.0 only reset when we press start again

4. Testing Table and Demonstration Pictures and Videos (25 pts):

The following points should be addressed in filling the two following tables and explanation of these test cases.

4.1. System Testing (10 pts)

4.1.1 System Modes Test

No.	Test cases	Test case description	Action	Units - Responsible	Expected Results	Actual results
0	Idling Mode (IM)	After the program is loaded, ensure the system enters idling mode after the defined inactivity period and resumes normal operation upon user interaction	Compile and upload the program	Module C: Network 1	After the specified inactivity period, conserving resources, and resumes normal functionality instantly when interacts with the system.	After the specified inactivity period, conserving resources, and resumes normal functionality instantly when interacts with the system.
1	Single Mode (SM) – Mode 1	Allows the system to execute a command only once at a time. After completion, the system must wait for the next command input before it can proceed, ensuring that each command is processed sequentially without overlap.	Press the Start button then turn the switch to the left which has a symbol "1" and turn it back.	Module C: Network 1, 2, 34 Module A: Network 3,4,5,6,18,19, 20 Module B: Network 7,8,9,10, 11 Module D: Network 21, 22, 23, 24, 25, 27, 28, 30, 31, 35	Once mode 1 is selected, the system processes only one workpiece at a time. After completing the sorting of that workpiece, it will wait for the next command before proceeding.	After successfully sorting a single workpiece, the system halts. To resume sorting in single mode, simply press the start button.
2	Continuous Mode (CM) – Mode 2	Allows the system to run continuously until one of the following events takes place: the stop button is pressed, the emergency button is triggered, or a fault is detected	Press the Start button then turn the switch to the right which has a symbol	Module C: Network 1, 2, 12, 13, 14, 15, 16, 17. Module A: Network 3, 4, 5, 6, 18, 19. 20. Module B:	Once mode 2 is selected, the system begins running and sorting continuously, stopping only when the stop button is pressed, the emergency button is	The system runs continuously, sorting all workpieces, and stops when one of the following scenarios occurs: the stop button is pressed, the emergency

			"II" and turn it back.	Network 7, 8, 9, 10, 11. Module D: Network 21, 22, 23, 24, 25, 26, 27, 28, 30, 31, 35	triggered, or a fault occurs (e.g. empty magazine, full slots, or system interference)	button is activated, or a fault is detected.
3	Triggering Stop Mode (STM)	Forcing the system to halt immediately and reset all operations.	Press the Stop button during the system cycle.	Module C: Network 1, 2. Module A, B, D: Network 16, 17	Once the Stop button is pressed, the whole system will be shut down and halt all operations immediately.	The system shut down and halt all operations and reset all the operations.
4	Triggering Fault Mode (FM)	Allows the system to stop automatically when any of the following scenarios occur: an empty magazine, full slots, or system interference.	Press the fault button on Module C	Module C: Network 12, 13, 14, 15	Once the Fault button is pressed, the system will immediately halt upon encountering any of the following scenarios: an empty magazine, full slots, or system interference	Pressing the Fault button once will cause the arm of station 2 to bounce to the middle Pressing the Fault button twice will activate the conveyor to move back to the left, returning to the home position. Pressing the Fault button three times will reset the entire system.
5	Exiting Fault Mode	Allow the system to exit Fault Mode after it has been triggered.	Press the Stop button after encounter the Fault Mode	Module C: Network 1, 2, 16	The system exits Fault mode, resets all operations, and is ready to start again at any time.	After Fault Mode is activated, press the Stop button to exit Fault Mode, reset all operations, and make the system ready to run again.

4.1.2 System Error Test

No.	Test cases	Test case description	Action	Unit Responsible	Expected Results	Actual results
1	Sort slots are full	Ensure the system stops when any of the three slots are full	Place all the workpieces into one slot to fill it or allow the system to run until one of the slots becomes full.	Module D Network: 26, 32	The system stops immediately after one of the three slots becomes full.	The system successfully stops immediately after one of the three slots becomes full.

2	Magazine empty	Ensure that the system halts automatically when the magazine is empty, preventing further operation until the magazine is replenished	Get out all the magazine or allow the system to run until there is no more magazine	Module A: Network 6, 7	When all the workpieces have been sorted and no magazines remain, the system will immediately stop after the last workpiece has been sorted	After the last workpiece has been sorted, the entire system will stop and remain inactive until more workpieces are loaded.
3	Swivel is stuck	Ensure the system stops if manual intervention is required	Press the Stop button when the Swivel is stuck	Module C: Network 1, 2, 16, 17	The Stop button should trigger appropriate recovery actions such as halting operations or attempting to reset the system.	Pressing the Stop button will trigger immediate stop, then can manually remove the obstruction or adjust the van. The system resumed normal operations after confirming the swivel was free to move
4	No piece delivered to slider tray	The module D operates the conveyor to move only if the workpiece is delivered to slider tray. It should not be operated if no workpiece is delivered to slider tray.	Open less the valve for the module B cylinder, so the workpiece fails to deliver to slider tray on the module D	Module B, Module D Network: 9, 21	Module D waits for the workpiece to be delivered, and Module B keep operates to deliver the workpiece from the module A to module D.	Module D waits for the workpiece to be delivered, and Module B operates to deliver the workpiece.
5	Start button is pressed when the cycle is not finished	Pressing start button during operation cycle does not affect the operation	Press start button during operation cycle either single cycle mode or continuous cycle mode	Module C Network: 1	Pressing only start button should not affect the cycle that is in progress	The cycle keeps going even though the start button is pressed
6	Emergency Button is pressed	Ensure the emergency button shut down all system immediately	Pressing emergency button has to stop the system and reset everything includes inputs, outputs, and memories	Module C: Network: 1, 7, 13, 14, 15, 16, 17, 18, 20, 21, 23, 24, 26, 27, 30, 32, 33	When the emergency button is pressed, the system stops the operation and reset all inputs, outputs, and memories.	When the emergency button is pressed, the system stops the operation and reset all inputs, outputs, and memories.

4.2. Solution Explanation (10 pts)

1. All sort slots are full.

"Sort slots are full" is designed to assess the system's behavior when any of the three slots reach full capacity. The procedure for this test involves either placing all workpieces into a single slot until it is filled or allowing the system to run until one of the slots is full. The unit responsible for this test is Module D. The expected outcome for this test case is that the system should halt immediately once any of the slots reaches its capacity. The actual results aligned with the expected behavior, as the system successfully stopped when one of the three slots became full. Then the operator can take out the workpiece and the system will start again after pressing the Start button.

2. Magazine is empty.

This test case focuses on the scenario in which the magazine responsible for storing workpieces becomes empty. The system must automatically halt when the magazine is depleted, preventing further operations until it is replenished. The system is required to either clear all workpieces or continue running until all are processed. Managed by Module A, the system will stop after processing the last workpiece and remain inactive until the magazine is refilled. This ensures that no unnecessary operations occur when the magazine is empty, preserving system efficiency and functionality.

3. Swivel is stuck.

This test case evaluates the system's response when the swivel mechanism becomes stuck. Pressing the Stop button should immediately halt the system, allowing for manual intervention to remove the obstruction or adjust the swivel. Module C is responsible for this action. The expected outcome is an instant system halt, enabling the operator to address the issue. Once the swivel is freed, the system should resume normal operations, ensuring a smooth and efficient recovery.

4. Start button is pressed when cycle have not completed.

This test case examines the system's response when the Start button is pressed during operation. The system should ensure that pressing the Start button does not disrupt the ongoing cycle. Module C is responsible for this action. The expected outcome is that the Start button has no effect, and the system continues its operation without interruption, preventing redundant inputs from interfering with the process.

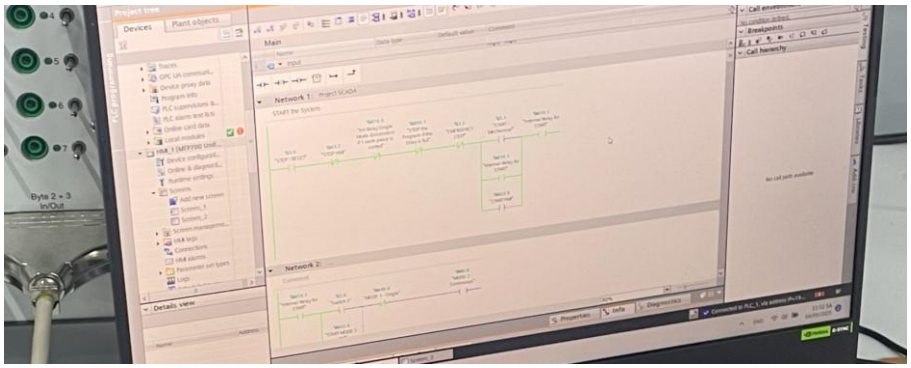
5. Emergency Button is pressed.

This test case evaluates the system's response when the emergency button is pressed. The goal is to ensure an immediate shutdown, overriding all ongoing operations to prevent further risks. Upon

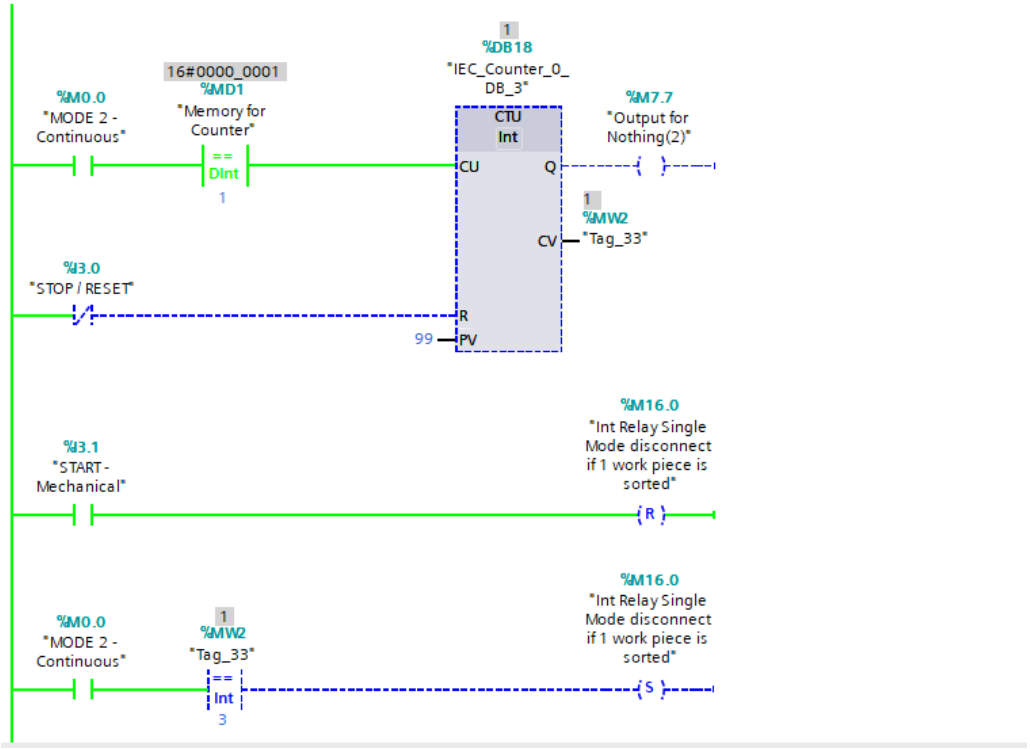
activation, the emergency button should stop all system activities without delay. Module C is responsible for executing this function. The expected outcome is a prompt halt of the system, ensuring a safe environment and minimizing any potential complications.

4.3. Demonstration Pictures and Videos (5 pts)

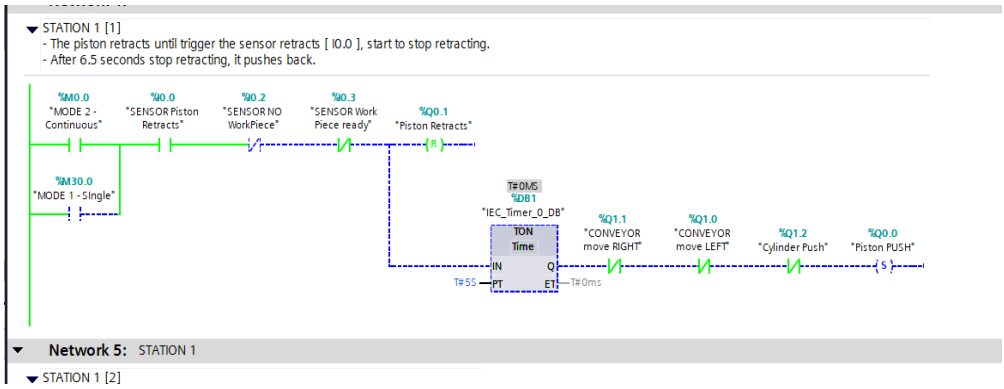
- Pictures
    - Start Button turns on when system is ready – Idling Mode



    - Fault Mode was triggered, whole system stops.
    - All sorting slots full, Fault Mode activated.



    - Magazine Empty, Fault Mode activated.



- Video link (Please access with your RMIT account)
  - Single mode: single mode
  - Continuous mode: continuous mode.
  - Trigger Stop Button: stopbutton.
  - Fault when magazine empty: fault when empty magazine.
  - Fault when all slots full: fault when magazine full.
  - HMI control: HMI control.

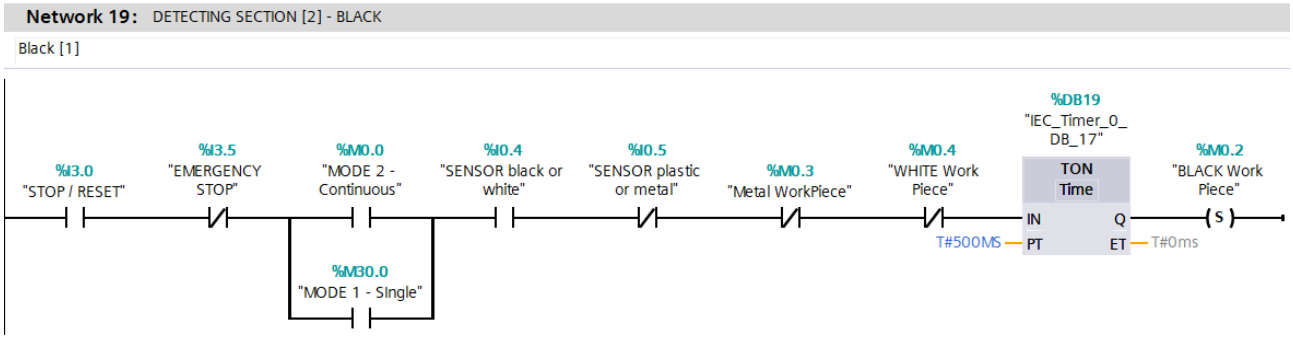
References

References: Including books, technical journals, patents, lectures. (if applicable). Please follow the referencing guide by our RMIT Learning Lab (link). You may also use the Easy Cite referencing tool by our library (link). Improperly cited sources will receive zero credit.

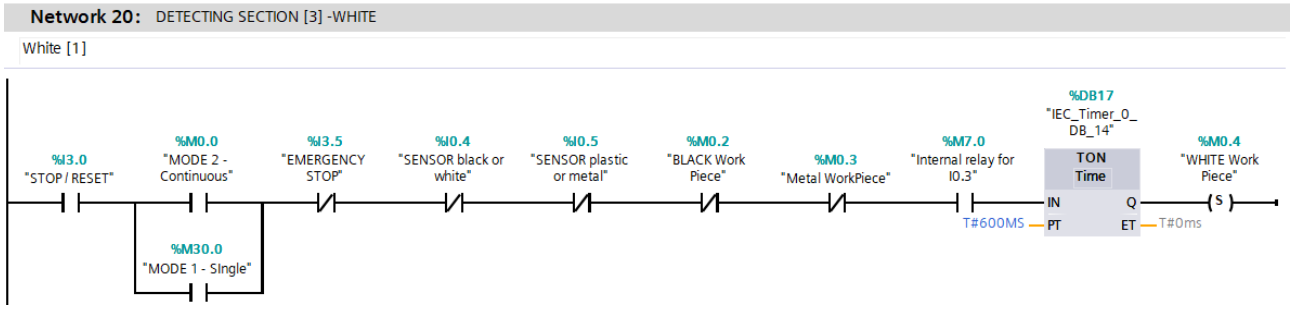
[1] E. Colgate, A. Bicchi, M. A. Peshkin and J. E. Colgate, "Safety for physical human-robot interaction," in *Springer handbook of robotics*, Springer, 2008, p. 1335–1348.

[2] S. Haddadin and E. Croft, "Physical human–robot interaction," *Springer handbook of robotics*, p. 1835–1874, 2016.

Appendix



Network 19 – detecting black workpiece



Network 20 – detecting white workpiece

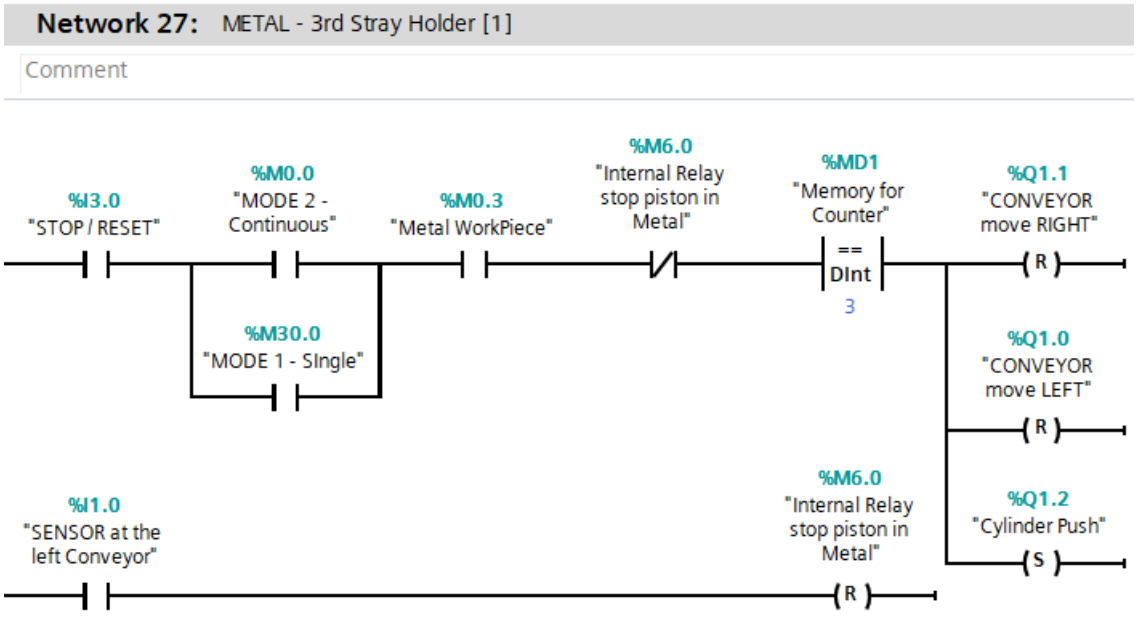


Figure : Network 27 – Sorting Metal [1]

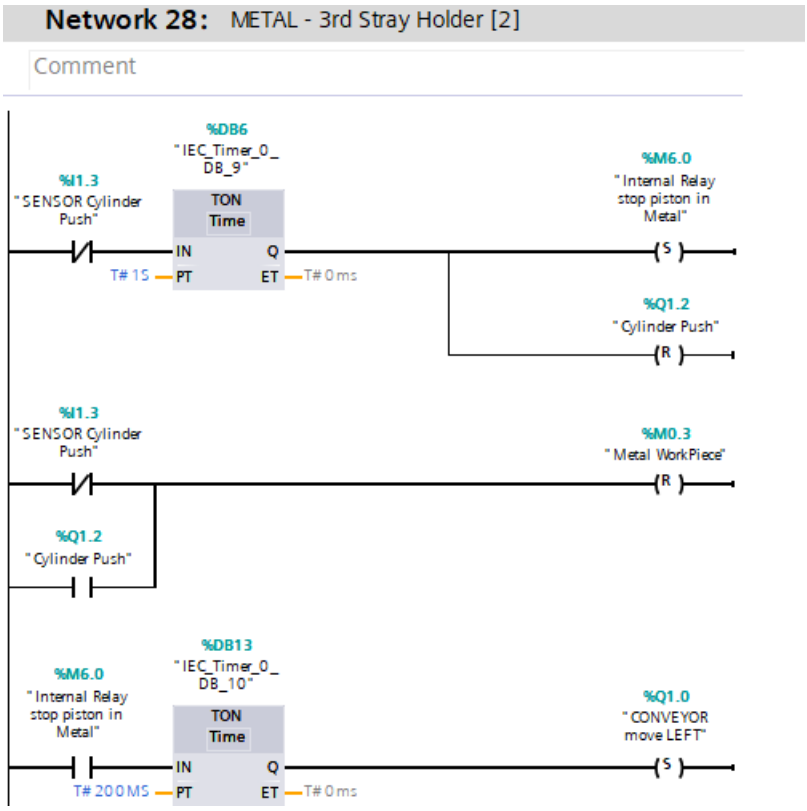


Figure : Network 28 – Sorting Metal [2]

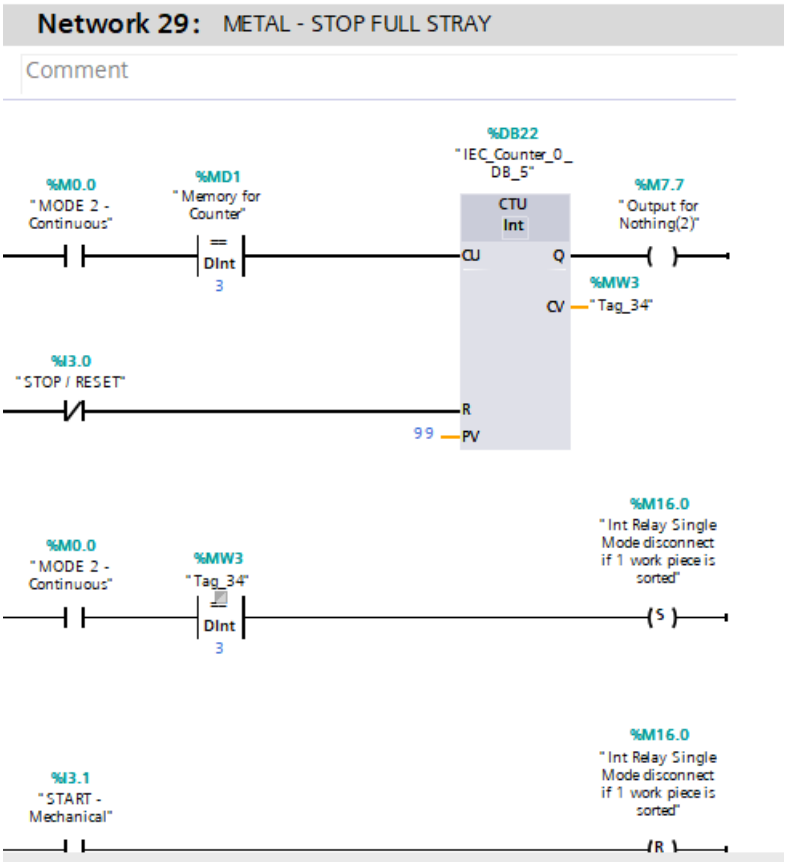


Figure : Network 29 – Stop Full Stray for Metal

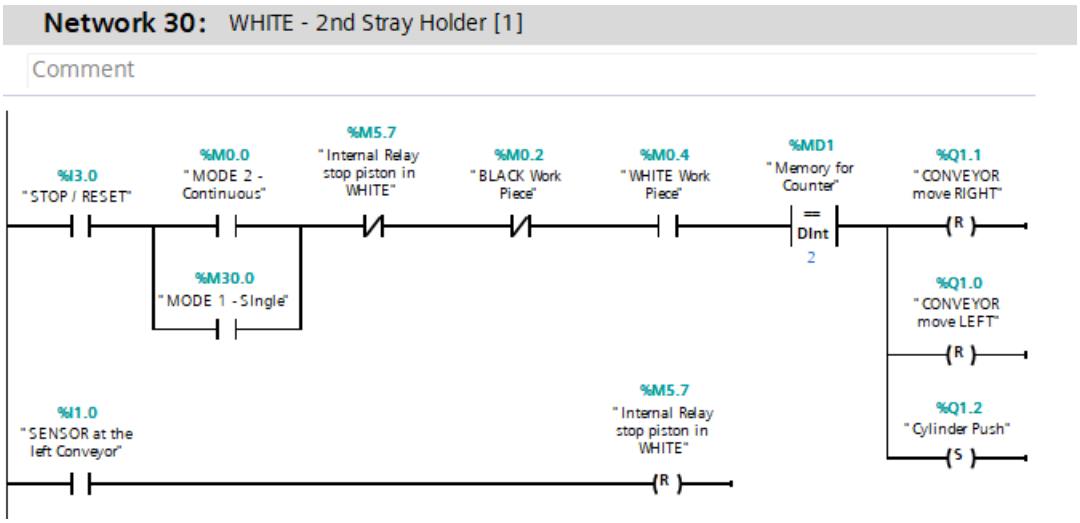


Figure : Network 30 – Sorting White [1]

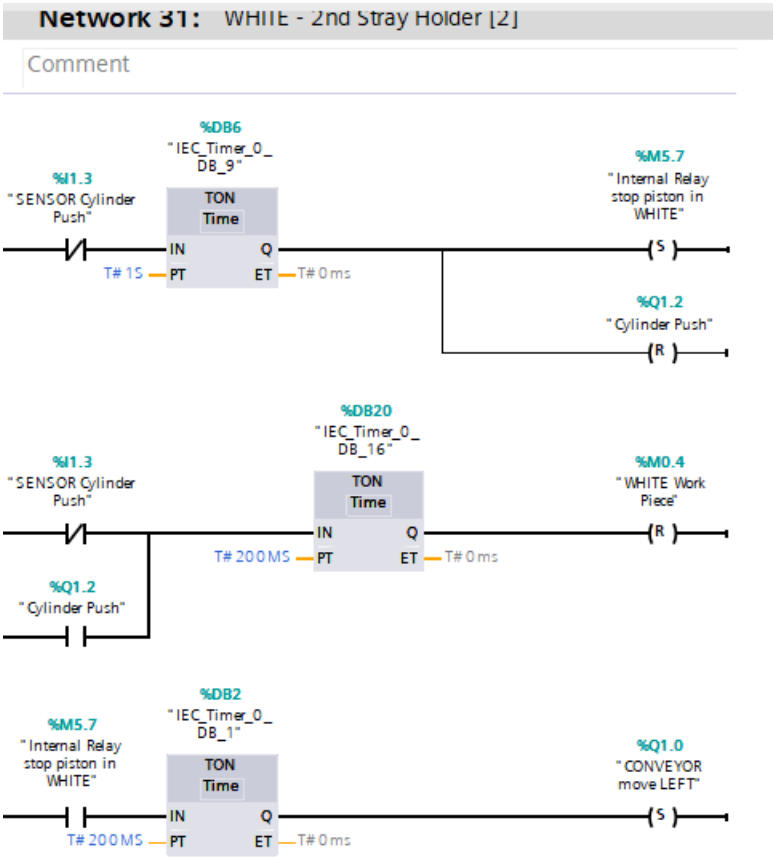


Figure : Network 31 – Sorting White [2]

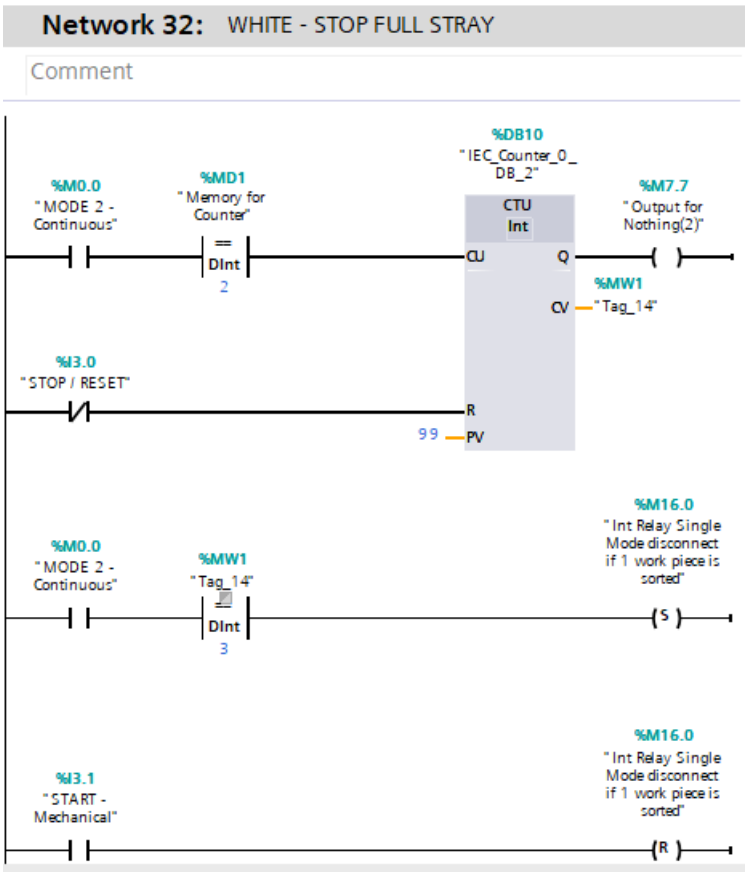


Figure : Network 32 – Stop full stray for White